

THE UNIVERSITY *of* EDINBURGH

Facility Location and Network Rebalancing with applications to Edinburgh's Cycle Scheme

by

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Executive Summary

In this report we propose an initial cycle scheme network of 40 nodes populated with 400 bikes, totalling an initial investment cost of £1.4 million. We estimate this scheme to pay back within 7 years using a pay-per-ride funding approach, advertising sponsorship and government subsidies. Our proposal is generated from a comprehensive software solution, motivated by the academic literature surrounding Stochastic Optimization, Location Science and Vehicle Routing.

Our report contributes to the existing academic literature in the following way. First, we propose a Markov Chain model which enables seasonality-aware demand scenarios. Second, we model and implement a two-stage stochastic MIP model for location of bike stations that utilizes the seasonal demand scenarios. Finally, we propose a rebalancing model in spirit of VRP-PD literature. We implement a MIP solution which solves instances of up to 40 nodes to optimality within 1000s. We further extend the existing literature by adapting the LKH-3 algorithm to solve for our model; enabling the solving of large instances, tested on up to 1000 nodes within a time limit of 1000s.

1 Introduction

Cycling is a convenient and time-efficient method of transport for short-range journeys; commuting to and from work; or for tourists seeking to explore a new city swiftly. Cycling is a more sustainable method of transportation compared to using cars or buses, and is correlated to lower levels of traffic congestion in urban areas [1]. In efforts to improve a city's sustainability, connectivity, and population wellbeing, it is no surprise that city councils across the UK are considering the introduction of cycle schemes. These schemes have shown to be effective in cities including London, Manchester and Leeds.

Edinburgh is planning to introduce a new cycle-hire scheme. This scheme will consist of a set of fixed bike station locations along with a set of electric-powered bicycles. A previous scheme operated between November 2018 and September 2021, however the scheme was not profitable. The primary concerns of the scheme were high levels of crime increased maintainability costs, lack of rebalancing, and reduced ridership. Furthermore, the scheme only covered 20% of Edinburgh's populated regions (these regions are referred by the Council as data zones). The Council seek our guidance to propose a new set of bike station locations that better service Edinburgh's population and a network rebalancing solution to improve the availability of bikes across the network.

In our report, we present both academic and business contributions. We contribute to the academic literature in the following way. We first propose two methods of generating demand scenarios; historical cycle scheme trip data aggregation followed by a Markov-chain model to account for 80% of data zones not captured by the previous cycle scheme. Second, we present a two-stage stochastic model for the bicycle station location problem with demand scenarios. We directly solve the model with commercial solvers, XPRESS and Gurobi. We additionally experiment with CPLEX's out-of-the-box Benders Decomposition strategy. Third, we present a rebalancing model motivated by the Pickup-Delivery Problem literature. We solve our model as a MIP using commercial solvers. To solve for instances exceeding 40 nodes, we extended an open-source implementation of the Lin-Kernighan-Helsgaun-3 (LKH-3) heuristic to solve our model, enabling solutions to all of our problem instances up to 1000 nodes. We implement our modelling and algorithmic proposals in a software solution to propose an initial bicycle network of 40 stations and 400 bikes that covers 200,000 historical journeys annually at an investment cost of £1.4 million.

The remainder of this report is organized as follows. In Sect. 2, we present the background on related works from the literature which motivate our project direction. In Sect. 3, we present our proposed modelling approaches. In Sect. 4, we propose our algorithmic approaches. In Sect. 5, we present our computational results, proposed Edinburgh bike station locations and funding strategy. Finally, in Sect. 6, we conclude our report and propose future research directions.

2 Background

In this section, we present a brief review of the related literature surrounding facility location, stochastic optimization and vehicle routing. These ideas influenced our modelling and algorithmic approaches, later explained in Sect. 3 and Sect. 4 respectively.

Facility location studies the problem of deciding the location of facilities, such as warehouses or factories. We refer the reader to [2] for a comprehensive review of the modelling and algorithmic approaches within the location science community. The most common formulations involve selecting a sub-set of facilities, p, from a discrete set of candidate facilities. The goal is to find the sub-set of facilities which minimizes instalment costs while maximizing some utility function. High costs are typically associated with facility establishment and removing facilities may incur sunk costs. This motivates the careful consideration of facility locations and evaluation of their utility under different scenarios. For public services, defining the utility function is not intuitive, and is often approximated by the use of surrogate measures. One surrogate measure is the amount of demand or geography covered by the locations, captured by the maximum covering location problem, first formulated in [3]. Suppose instead one wishes to choose p facilities to minimize the maximum distance from any demand point to its nearest facility. This is referred to as the p-center problem. Alternatively, one may consider finding p facilities to minimize the total weighted distance from all demand points to their nearest facility. This is referred to as the p-median problem.

In stochastic optimization, the uncertainties are assumed to be modelled by a known distribution. Two-stage stochastic programming is one of the most famous approaches, typically optimizing over a set of finite scenarios, each with known probability of occurring. In this framework, the problem is modelled by a set of first-stage, here and now, decisions, and a set of secondary decisions which model the behaviour under realisation of uncertainty, referred to as wait-and-see decisions. Benders decomposition, proposed by Jacques Benders in 1962 [4], is one of the most famous approaches for solving stochastic programming problems. One of the first application of Benders to stochastic programming was by Van Slyke and Wets in 1969 [5]. In classical Benders, the master problem contains integer variables only and the sub-problem are continuous variables only. The use of continuous variables only in the benders sub-problem enables generation of benders optimality and feasibility cuts from the duals. In stochastic programming problems, one can solve all scenarios in a single sub-problem, referred to as Single-Cut Benders. However, to better exploit the independence of scenarios, one Benders sub-problem per scenario can be formed. Each sub-problem can therefore be solved more efficiently than the scenario-aggregated sub-problem. Furthermore, in each Benders iteration, this approach enables a Benders cut to be added to the master problem for each scenario rather than a single cut for all scenarios. This is referred to as Multi-Benders. We restrict our report to use a classical single-cut Benders approach for simplicity and identify the adoption of multi-cut Benders techniques as a direction for future study.

The Travelling Salesman Problem (TSP) involves a single salesman who must visit a set of cities exactly once and return to the starting city. The goal is to find the shortest possible route that. TSP is a well-known NP-hard problem. The Multiple Travelling Salesman Problem (M-TSP) is an extension of TSP to M salesmen. Each city must be visited exactly once, and all salesmen must return to the starting city. M-TSP can be regarded as a special case of the broader Vehicle Routing Problem (VRP). In VRP, given a fleet of vehicles and one or more depots, the goal is to determine the optimal routes that allow all nodes to be visited while minimizing the total distance travelled. Pickup and Delivery Problems (PDPs) are variants of VRP that model additional constraints. For example, certain items must be picked up at designated nodes and delivered to specific destinations. In these problems, a fleet of capacitated vehicles performs sequences of pickups and deliveries to achieve a balanced system state while minimizing operational costs. The Bike Rebalancing Problem (BRP) is a variant of PDP, where a fleet of vehicles must rebalance the stock of a single commodity, bikes, from one set of nodes of surplus (pickups) to nodes with deficits (deliveries). An outline of the basic variants of the BRP and their characteristics are presented in [6]. In our work, we focus on the static case of BRP, where it is assumed that the demand does not change while the system is being rebalanced.

Due to the computational challenge of VRP and its variants, heuristic approaches are often used for solving real-world problems. These include, local-search, variable neighbourhood search and simulated annealing. The Lin-Kernighan-Helsgaun (LKH) algorithm [7] is one of the state-of-the-art (SoTA) local search algorithms for the TSP, and LKH-3 is an extension which solves variants of TSP such as VRP with Time Windows (VRPTW) and PDPs. LKH-3 works in the following way. First, the algorithm transforms an M-TSP into TSP [8]. Further, asymmetric TSP instances are transformed to an equivalent symmetric one [9]. Constraints in LKH-3 are handled through a penalty function. A penalty function is a function which maps a tour τ to a cost pair (P, C), where P is the constraint penalty and C is the tour cost. The objective of LKH-3 is to first minimize constraint penalty costs, P (prioritizing feasibility); and second minimize tour cost, C. To improve solutions, LKH-3 leverages variable K-Opt moves. A k-Opt move is a procedure which improves a tour by removing k edges and adding k new edges to create a new tour of reduced cost. The process is repeated until no further improvement can be made, resulting in a locally optimal tour. In our work, we adapt the penalty function of LKH-3 to solve our BRP model.

3 Modelling Approach

In this section, we present our modelling approaches. In 3.1, we present our scenario generation models. We present the formulations of our facility location model in 3.2 and our rebalancing model in 3.3.

3.1 Scenario Generation

We define a scenario as a set of demand nodes. Each demand node has a distinct inbound and outbound demand corresponding to the number of journeys arriving at or leaving from the demand node respectively over a time period. We choose demand nodes to be data zones in the Scottish Index of Multiple Deprivation

2020 dataset (SIMD) [10]. We do this because data zones in SIMD are enriched with socio-economic data and are regionalized. Specifically we use the population of each data zone. We propose two approaches for modelling demand. The first is to aggregate historical trips over the 4 years that the scheme previously ran. From this we incorporate hourly, daily and seasonal trends in our demand modelling. In our second approach we use a Markov chain simulation to model the population flow between data zones based on population of the data zones and their proximity. The Markov approach predicts demand not serviced by the previous scheme.

3.1.1 From historical journeys

We present a brief analysis on the historical trip data of the previous scheme. In Figure 1, we show the aggregation of outbound trips by hour of day and day of week. We observe distinct patterns of behaviour during weekdays, and weekends. Peaks are correlated to user types. For example, the weekday morning peaks are highly correlated to the typical working hours of office-based jobs of 9am-5pm, and similarly the lecture schedules of students. This is further supported by no 8am peak on the weekend data, when most office-based professionals are not working and as there are no lectures on weekends. Finally, we suspect the high usage of the scheme on weekends between 10am-5pm, and 1pm peak may be motivated by tourism and leisure activities. Through the modelling of seasonality, bike station location and rebalancing decisions can be made to incentivize sufficient stock of bikes in the right geographical areas to service spikes in demand; and implicit consideration of fairness across the user types we mentioned.

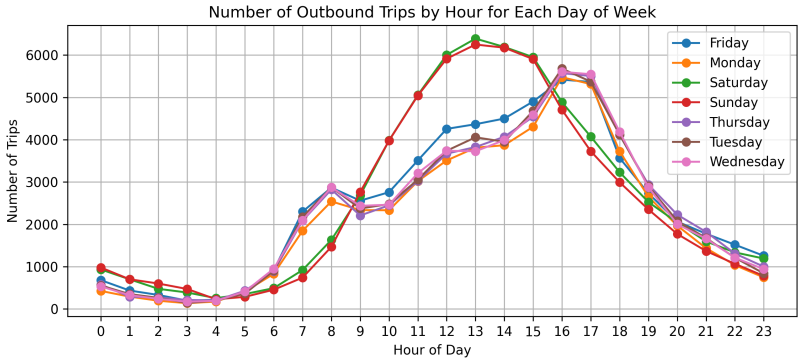


Figure 1: Historical outbound trips per hour by day of week on Edinburgh's previous cycle scheme.

Based on the historical journeys we create scenarios for mean number of daily journeys split on Spring, Summer, Winter, Autumn, Weekday vs weekend, and Time of day (morning, midday and afternoon), totalling 24 scenarios. Using the split, we compute the inbound and outbound journeys for each cycle location, and map inbound and outbound demand to the nearest data zone; these data zones are the demand nodes. Each demand node is mapped to candidate locations that are in close enough proximity to service the node.

3.1.2 Markov chain based on population and distance

Markov chains provide a framework for modelling the evolution of systems; with transition probabilities between states that direct how the system evolves. The use of Markov chains is well established in the literature, ranging from transport networks [11] to community detection in biomolecules [12]. In this research, we model the flow of population (journeys) in Edinburgh as a discrete-time Markov chain.

We construct a bidirectional graph, each distinct data zone is a distinct node in the graph, and each vertex model a population flow between any two nodes; governed by a weight. We perform a random walk on the graph where transition probabilities are determined by population and proximity of nodes. S is the full state space of the system: all populations across all nodes for all discrete time steps, and is given by the Cartesian product of S_T and S_N shown in Eq.1. S_T is the state space of discrete time steps in the simulation from $t = 0$ up to $t = T$, with steps of 1, given by Eq.2. S_N is the state space containing the total population of each node at a single time step, given by Eq.3. Furthermore, the simulation satisfies the Markov property (Eq.4); that is that the next state of the system j only depends on the current state i and not the states before it.

$$S = S_T \times S_N \tag{1}$$

$$S_{\mathrm{T}} = \{0, 1, \ldots, T\} \tag{2}$$

$$S_N = \{s_i \mid i \in D\}, \quad s_i \text{ is the population in node } i \tag{3}$$

$$P(S_{t+1,N} = j \mid S_{t,N} = i, S_{t-1,N}, \ldots, S_0) = P(S_{t+1,N} = j \mid S_{t,N} = i). \tag{4}$$

The initial state $S_{0,N}$ is given by the population of each data zone in the dataset. d_{ij} is the haversine distance between any two data zones i and j. n_i, n_j are the total populations of data zones i and j respectively. The weights connecting any two data zones i and j is given by Eq.5. Note that self transitions, i.e $i = j$ are not allowed, and these weights are set to zero.

$$w_{ij} = \begin{cases} \dfrac{n_i n_j}{d_{ij}}, & i \neq j, \\ 0, & i = j, \end{cases} \qquad i, j \in D \tag{5}$$

In Eq. 6, we present the transition matrix, containing the transition probabilities; each specific transition probability from a node $i' \in D$ to node $j' \in D$ is given by Eq.7. Note that every node is connected to every other node, and the sum of transition probabilities from any node add up to 1. We assume a node cannot transition to itself, hence transition weights and probabilities are zero when $i = j$.

$$P_{ij} = p(i, j) \tag{6}$$

$$p(i', j') = \frac{w_{i'j'}}{\sum_{j \in D, j \neq i'} w_{i'j}} \tag{7}$$

3.2 Bike Station Location Model

In this section we present our proposed mathematical model for solving the bike station location problem. We define a set of candidate bike station locations and customer demand nodes, by $i \in I$ and $j \in J$ respectively. Our model decides on which bike stations to open in order to maximize demand coverage, weighted across a set of demand scenarios. Define $x_i \in X$ as the binary variable to open bike station i (when 1) or not (when 0). At most β bike stations can be constructed. We define a set of demand scenarios as $s \in S$, each with probability $p_s \in P$ of occurring. Each customer demand node has a defined inbound (IB) and outbound (OB) demand to service per demand scenario, denoted by $d^{IB}_{j,s}, d^{OB}_{j,s} \in D$ respectively. We model the distinction between IB and OB demand as we observed asymmetric demand flow was during the day on historical trip data. Define $y^{IB}_{i,s}, y^{OB}_{i,s} \in Y$ as the continuous variables denoting the amount of inbound and outbound demand respectively to allocate from demand node j to bike station i in scenario s. We chose to approximate demand allocation with continuous rather than integer variables as continuous variables results in a simpler second-stage problem that is more tractable and we assume a continuous approximation suffices for long-term planning. In Table 1, we summarise our decision variables.

Table 1: Bike Station Location Problem variables

Notation	**Description**
$x_i \in X$	The binary decision to open i. 1=open, 0=closed.
$y^{IB}_{i,j,s} \in Y$	The allocated inbound demand from demand node j to bike station i in scenario s
$y^{OB}_{i,j,s} \in Y$	The allocated outbound demand from demand node j to bike station i in scenario s.

Our objective function is to minimize the fixed cost of installing docking stations (here and now decisions), and maximize, per scenario, the probability of the scenario occurring multiplied by the demand allocation in the scenario (second-stage decisions). Demand coverage is not enforced as a hard constraint as this is susceptible to model infeasibilities from model inputs.

$$\min \quad \underbrace{\sum_{i \in I} f x_i}_{\text{Station opening cost}} \quad -(\quad \underbrace{\sum_{s \in S} p_s \sum_{i \in I} \sum_{j \in J} y^{IB}_{i,j,s} + y^{OB}_{i,j,s}}_{\text{IB and OB Demand coverage across scenarios.}} \quad) \tag{8}$$

Constraint 9 enforces a maximum budget (user parameter), β, on the number of bike station locations.

$$\sum_{i \in N} x_i \leq \beta \tag{9}$$

Constraints 10 enforce that, per scenario, the total IB demand serviced by a bike location cannot exceed a maximum limit (user parameter), $\alpha_{i,s} \in A$.

$$\sum_{j \in J_i} y^{IB}_{i,j,s} \leq \alpha_{i,s} x_i \quad \forall i \in I, \forall S \in S \tag{10}$$

We assume that, in a given scenario, bike rebalancing only occurs due to customer usage. In this spirit, constraints 11 ensure that the total inbound demand minus outbound demand, and the total outbound demand minus inbound demand cannot exceed the capacity, c_i, of docking station i, per scenario.

$$\sum_{j \in J_i} y^{IB}_{i,j,s} - y^{OB}_{i,j,s} \leq c_i x_i \quad \forall i \in I, \forall S \in S \qquad \sum_{j \in J_i} y^{OB}_{i,j,s} - y^{IB}_{i,j,s} \leq c_i x_i \quad \forall i \in I, \forall S \in S \tag{11}$$

Constraints 12 enforce that for each scenario, respectively, the total IB and OB demand allocated from demand node j cannot exceed the total demand of the node. Total IB and OB demands of demand node j are demoted by $d^{IB}_{j,s}$ and $d^{OB}_{j,s}$ respectively.

$$\sum_{i \in I_j} y^{IB}_{i,j,s} \leq d^{IB}_{j,s} \quad \forall j \in J, \forall s \in S \qquad \sum_{i \in I_j} y^{OB}_{i,j,s} \leq d^{OB}_{j,s} \quad \forall j \in J, \forall s \in S \tag{12}$$

Constraints 13 enforce demand assignment from j to i for inbound and outbound demand in each scenario is non-negative. Finally, 14 models that picking a station location is a binary decision.

$$y^{IB}_{i,j,s} \geq 0, \quad y^{OB}_{i,j,s} \geq 0 \qquad \forall i \in I, \forall j \in I, \forall s \in S \tag{13}$$

$$x_i \in \{0, 1\} \qquad \forall i \in I \tag{14}$$

3.3 Rebalancing Model

In this section we present our proposed mathematical model for solving the bike rebalancing problem. Let G be a directed graph with nodes $i \in I$ and edges $(i, j) \in E$. A set of vehicles, $v \in V$, may traverse this graph. Define $t_{i,j,v} \in T$ as the transit time from node i to j using vehicle v. We assume transit times are asymmetric between nodes i and j. Define τ as max allowed transit time per vehicle tour. Define c as the cost of travelling per unit of transit time. Each node i, has a number of bikes away from the target bike stock level, referred to as $\gamma_i \in \Gamma$. A positive γ_i implies node i has a surplus of γ_i bikes. A negative γ_i implies node i has a deficit of $-\gamma_i$ bikes. Finally, we weight rebalancing in the model with a user parameter, β.

$x_{i,j,v} \in X$ are binary variables indicating vehicle v travels from location i to j when 1 and else 0. $f_{i,j,v} \in F$ denotes the integer flow of bikes from location i to j using vehicle v. $s_i \in S$ denotes the integer number of bikes location i consumes (when positive) or releases (when negative) in the rebalancing process. Finally, $u_{i,v} \in U$ are continuous variables introduced to model the Miller-Tucker-Zemlin (MTZ) constraints that eliminate sub-tours. [13]. Subsequently, model decision variables are presented in Table 2. We use a 3-index formulation of VRP which explicitly indexes each vehicle rather than the 2-index formulation that does not. Though a 2-index formulation is more efficient on commercial solves than a 3-index one (fewer symmetric solutions), we opt for a 3-index formulation due to its increased flexibility/extensibility and is simpler to implement within our project timelines.

Table 2: VRP Problem variables

Notation	Description
$x_{i,j,v} \in X$	Binary variable indicating vehicle v travels from location i to j when 1, and not when 0.
$f_{i,j,v} \in F$	Integer flow variable indicating the number of bikes vehicle v transports along $(i,j) \in E$.
$s_i \in S$	Integer number of bikes i consumes (when +) or releases (when -) in re-balancing.
$u_{i,v} \in U$	Continuous auxiliary variable for the MTZ constraints for node i and vehicle v.

Our objective function is to minimize the total transit time travelled by all vehicles plus the incentive for re-balancing of nodes with surplus (s_i is negative in such case) minus the incentive of re-balancing of nodes with deficit (s_i is non-negative in such case).

$$\min \underbrace{\sum_{i \in I} \sum_{j \in I} \sum_{v \in V} t_{i,j,v} x_{i,j,v}}_{\text{Transit time cost}} + \underbrace{\sum_{i \in I^-} \beta s_i - \sum_{i \in I^+} \beta s_i}_{\text{Re-balancing incentive}} \tag{15}$$

In 16, we model flow conservation of vehicles: Per node and vehicle, the number of vehicles leaving node i minus the number of vehicles entering node i equals zero.

$$\sum_{j \in I} x_{i,j,v} - \sum_{j \in I} x_{j,i,v} = 0 \qquad \forall i \in I, \forall v \in V \tag{16}$$

In 17, we model that each node must be visited once, except for the depot. In future, one could consider relaxing the requirement to visit every node, in spirit of Prize-Collecting VRP.

$$\sum_{j \in I \setminus \{i\}} \sum_{v \in V} x_{i,j,v} = 1 \qquad \forall i \in I \setminus \{0\} \tag{17}$$

In 18, we enforce that each vehicle must leave and enter the depot once respectively.

$$\sum_{j \in I \setminus \{0\}} x_{0,j,v} = 1 \qquad \sum_{i \in I \setminus \{0\}} x_{i,0,v} = 1 \qquad \forall v \in V \tag{18}$$

In 19, we leverage the Miller–Tucker–Zemlin (MTZ) constraints, adapted for VRP [13] [14]. This eliminates sub-tours by enforcing an ordering on the visiting of each node. We define $M = |I \setminus \{0\}|$, the number of all locations except the depot. This formulation results in a weak LP relaxation, nevertheless, we opt for this approach due to its simplicity and uses polynomial number of variables and constraints - which is practical for constructing the MIP.

$$u_{i,v} - u_{j,v} + M x_{i,j,v} \leq M - 1 \qquad \forall i \in I \setminus \{0\}, \forall j \in I \setminus \{0\}, \forall v \in V \tag{19}$$

In 20, we enforce maximum transit time restrictions per tour of τ. Tours with excessive transit times may not allow for appropriate re-balancing of nodes before the next wave of demand.

$$\sum_{i \in I} \sum_{j \in I} t_{i,j,v} x_{i,j,v} \leq \tau \qquad \forall v \in V \tag{20}$$

Constraint 21 enforces bicycle re-balancing demand flowing through an edge (i,j) on vehicle v is bound by the total truck capacity on the edge. $c_v \in C$ is the capacity of vehicle v.

$$f_{i,j,v} \leq c_v x_{i,j,v} \qquad \forall i \in I, \forall j \in I \setminus \{i\}, \forall v \in V \tag{21}$$

Constraint 22 models flow conservation of bikes across the tours. The total flow of bikes into a node minus the flow out of the node plus the flow consumed by the node must be equal zero. Since we assume atmost one vehicle can enter a node, we aggregate the flow constraints by vehicles.

$$\sum_{j \in J \setminus \{i\}} \sum_{v \in V} f_{j,i,v} - \sum_{j \in J \setminus \{i\}} \sum_{v \in V} f_{i,j,v} - s_i = 0 \qquad \forall i \in I \tag{22}$$

Let $i \in I^+$ and $i \in I^-$ denote the set of bike location nodes with a surplus and deficit of bikes respectively. Constraints 23 models adherence to stock re-balancing targets. On the left, we enforce that nodes who have

a surplus of bikes must donate between 0 and γ_i bikes to another node (and hence must receive between 0 and $-\gamma_i$ bikes). On the right, we enforce that nodes who have a bike deficit must receive between 0 and γ_i bikes from other nodes.

$$0 \geq s_i \geq -\gamma_i \quad \forall i \in I^+ \qquad 0 \leq s_i \leq \gamma_i \qquad \forall i \in I^- \tag{23}$$

Finally, in 24 and 25, we present our non-negativity and binary decision constraints respectively.

$$u_{i,v} \geq 0 \quad \forall i \in I, \forall v \in V \qquad f_{i,j,v} \geq 0 \quad \forall i \in I, \forall j \in J, \forall v \in V, \tag{24}$$

$$x_{i,j,v} \in \{0,1\} \qquad \forall i \in I, \forall j \in I, \forall v \in V \tag{25}$$

4 Algorithmic Approach

In this section, we present our algorithmic approaches for bike station location and bike rebalancing. The architecture of our software is presented in Figure 2. Based on our experimental results, presented later in Sect 5, we propose to use the modules highlighted in green.

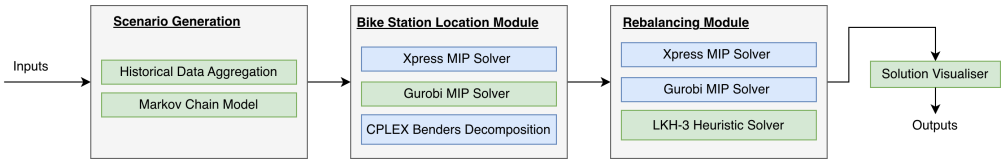


Figure 2: The end-to-end software solution architecture.

4.1 Bike Station Location Module

We solve the model presented in Sect 3.2 directly using Xpress, and Gurobi. We opted for this exact approach, rather than heuristics, in the spirit of seeking optimal solutions. Xpress was chosen as this is the solver recommended to use within the course. We additionally benchmarked with Gurobi as this solver is well-known to solve complex problems fast. Finally, as discussed in Sect. 2, this model can be reformulated as a classical benders decomposition. We leverage CPLEX's built-in support for Benders Decomposition which reformulates and solves the problem for us, out-of-the-box, using a Single-Cut Benders approach. Our Master problem consists of the binary candidate location decision variables and budget constraint with objective to minimize station opening cost plus sub-problem cost. Our sub-problem contains all continuous flow variables and flow-based constraints and has the objective to maximize covered demand, weighted across all scenarios.

4.2 Bike Rebalancing Module

Our rebalancing module leverages transit time data from Open Source Routing Machine (OSRM). We setup a local OSRM server and query this within our software solution for each edge in our network. In doing so, we enable realistic approximation of vehicle transit times. We enable a set of user-configurable parameters: Number of vehicles, vehicle bike capacity, cost per transit time, maximum transit time per tour, and rebalancing levels. Finally, the user inputs rebalancing stock levels per node.

We first solve the mathematical model in Sect. 3.3 using Xpress and Gurobi. Commercial solver selection is as per discussed prior in Sect. 4.1. Furthermore, we opt for this exact approach to enable optimality guarantees of our solutions. However, as per the literature and as discussed later in Sect 5, this approach does not scale well with the number of nodes and vehicles. We therefore propose a heuristic approach which enables scaling to large instances. Specifically, we extend the existing LKH-3 C-based implementation with a new penalty function. The penalty function is defined as the total sum of absolute constraint violations and amount of rebalancing which could not occur. The cost function is defined as minimizing the total transit time of each tour, which is unchanged compared to the original LKH-3 implementation. We interact with the adapted LKH-3 executable through a custom Python interface that we implemented. This interface generates the problem instances in the required LKH-3 input format and extracts the solution from the outputs.

5 Analysis and Results

In this section, we present our analysis and results. We show an example scenario from our Markov model, along with analysis on our algorithmic approaches for bike station location and network rebalancing. Finally, we propose an initial bike scheme consisting of 40 bike station locations. All experiments are conducted on a Mac M1 Pro machine with 16Gb of RAM. Xpress and Gurobi results leverage default parameter setup. Model files are published to our GitHub repository for reproducibility purposes. Xpress and Gurobi results leverage default parameter configurations.

Using our Markov model, proposed in Sect. 3.1.2, we generated scenarios for inbound and outbound demand. In Figure 3, we present an example scenario during summer.

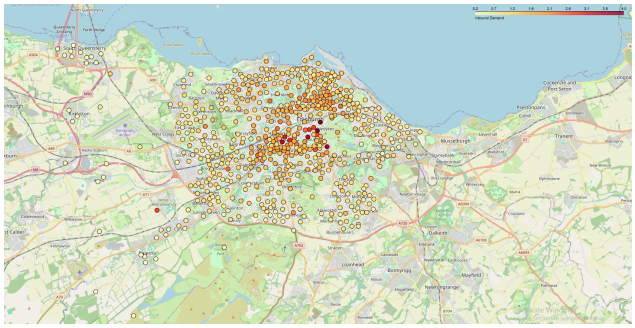


Figure 3: Map of Edinburgh with data zones circled showing demand per data zone, normalized to 750 trips.

Our Markov model predicts higher demand on nodes around the city centre, where the population is higher and data zones are closer in proximity. Data zones outside of Edinburgh city centre are further apart and have lower populations, giving a lower demand prediction. The previous scheme only serviced 104 data zones, which is reflected in our historical demand scenarios. Our proposed Markov model predicts demand for all 598 data zones; which increases accessibility of the scheme by up to 494 more data zones.

We constructed a bike location instance comprising of over 30,0000 candidate station locations, and 25 demand scenarios; weighted proportionately by the time duration. All algorithmic approaches solved our instance to optimality. Gurobi solved the instance within 21.4 seconds to optimality. This is likely explained by Gurobi's powerful heuristics approaches which found good initial solutions quickly. Xpress solved the instance to optimality in 186 seconds. Finally, CPLEX with Single Cut-Benders solved our instance in 1022 seconds which was approximately 10% faster than CPLEX under default configurations which solved to optimality in 1140 seconds.

In Table 3, we present scalability results on our proposed solution approaches for solving our proposed bike rebalancing model. Visualized solutions are presented in Appendix. A. To generate each instance, bike station location nodes are selected at random. Further, we set the bike rebalancing term to be dominating in the objective which results in 100% rebalancing on all of our experimentation instances. Therefore, we omit that information from the results table.

Nodes	Vans	Xpress MIP		Gurobi MIP		Adapted LKH-3 Heuristic	
		Transit Time [hrs]	Solve Time [s]	Transit Time [hrs]	Solve Time [s]	Transit Time [hrs]	Solve Time [s]
10	2	1.5	0.13	1.5	0.04	1.7	5.7
20	2	2.0	1.1	2.0	0.16	2.2	108
30	2	2.8	71	2.8	18	2.8	151
40	2	3.3	935	3.3	100	3.3	413
50	3	N/A	1000	N/A	1000	3.8	782
100	5	N/A	1000	N/A	1000	6.9	1000
1000	40	N/A	1000	N/A	1000	29.2	1000

Table 3: Scalability results of algorithmic approaches for solving our proposed bike rebalancing model.

In our experiments, both Xpress and Gurobi solved the 10-40 node instances to optimality within the time

limit. However, neither could even find a feasible solution to any of our instances beyond 40 nodes. The 50-node instance has over 15 thousand variables and constraints. The 1000-node instance has over 10 million variables and over 10 million constraints. The adapted LKH-3 heuristic found near-optimal solutions to the 10 and 20 node instance, optimal solutions for the 30, 40 and 50 node instance; and best known feasible solutions for the 100 and 10000 node instances. From our results, we conclude that on our instances up to 40 nodes, Gurobi is the superior choice compared to Xpress MIP and the Adapted LKH-3 approaches. Beyond 40 nodes, LKH-3 is the only approach in which feasible solutions are generated within the time limit. In order to strengthen our conclusions on scalability of approaches, parameter tuning of Xpress and Gurobi should be explored and experiments should extend to a more comprehensive number of instances of varying complexity and compositions in node locations, number of vehicles, rebalancing levels, and maximum tour transit times. If time would have permitted, we would have done so. In future, we would like to also assess the performance of our adapted LKH-3 heuristic on larger instances and assess the performance on other cycle schemes, such as London, which has over 800 cycle stations.

5.1 Proposed Bike Network

We propose an initial bike network with 40 station locations as presented with blue circles in Figure 4. This network is generated from our end-to-end software solution. The construction of 40 bike station locations will cost £1.2 million in instalments. Existing schemes around the UK have between 10-20 times more bikes than stations. Motivated by this range, and consider our scheme is new, we propose an initial investment of 400 bikes (10 per station on average), and propose to later evaluate if further bikes are required based on execution data of the new scheme. Assuming £500 per bike, this initial investment will cost £200, 000.

We additionally present a rebalancing simulation in a worst-case scenario in Figure 4. This provides a worst-case bound on the vehicle rebalancing cost. Our worst-case scenario assumes the following: Use of two vehicles, a maximum transit time per vehicle of 2 hours, every node must be visited, 50% of nodes must receive 5 bikes, and the remaining nodes must donate 5 bikes. The total transit time per of the routes is 3.11 hours, with one vehicle taking 1.79 hours and the other taking 1.32 hours. Translating to distances and assuming fuel cost of 15p per mile, the annualized fuel cost is £5,0000. Furthermore, assume labour costs must be paid hourly, and hence 4 hours of labour is required per day. Assume labour cost at £12 per hour. Annualized labour cost is estimated at £17,520. Therefore, annualized total rebalancing cost is estimated at £22,520.



Figure 4: Our proposed network location of 40 nodes with a worst-case scenario rebalancing route.

We propose to fund the scheme with the following business strategy, in priority order. Firstly, we propose a pay-per-ride model. Based on historical scheme coverage, our proposed network is estimated to serve 200, 000 cycle journeys yearly. Assuming a pricing strategy of £1 per ride, that equates to £200, 000 in annualized revenue. Assuming constant ridership and no maintenance cost, this funding approach will provide a return on the initial £1.4 million investment after 7 years of operation. This price-per-ride is competitive to other forms

of transport within Edinburgh. For example, a single bus journey costs £2.20. Second, we propose to offer advertisement on our bikes in exchange for the payment of maintenance and worst-case rebalancing cost of £22, 520 per year. Finally, we propose to consult with the government to negotiate subsidies. The effectiveness of this depends on the government's willingness to invest in CO_2 emission reductions in Edinburgh. Based on historical trips and a ridership of 200, 000 per year, the scheme is estimated to save 250 tonnes in CO_2 emissions through the reduction of 100,000 car journeys.

6 Conclusions and Future Works

In conclusion, we propose a comprehensive end-to-end software solution for demand prediction, bike station location and network rebalancing of bikes for Edinburgh's new cycle scheme. We contribute to the existing literature in the following way. First, we propose a custom Markov Chain model which enables seasonality-aware demand scenarios. Second, we model and implement a two-stage stochastic MIP model for location of bike stations that utilizes the seasonal demand scenarios. Finally, we propose a rebalancing model in spirit of VRP-PD literature. Our modelling approach is novel compared to the existing literature in that we model bikes as fungible demand - bikes picked up from any node can be dropped at any other node. We implement a MIP solution which solves instances of up to 40 nodes to optimality within 1000s. We further extend the existing literature by adapting the LKH-3 algorithm to solve for our model; enabling the solving of large instances, tested on up to 1000 nodes within a time limit of 1000s.

We leverage our framework to propose an initial cycle scheme network of 40 nodes with 400 bikes initially, totalling an initial investment cost of £1.4 million. We estimate this scheme to service 200,000 trips per year from historical data and simulated demand. We subsequently proposed a pricing and funding model which will enable the scheme to be profitable after 7 years. First, we propose a pricing of £1 per trip. This improves transport affordability in Edinburgh, costing the user less than half the price of a single bus journey. Second, we propose to offer advertising sponsorship in order to cover maintenance and network rebalancing costs. Finally, we propose to consult with the government to negotiate subsidies. The effectiveness of this will on the government's willingness to invest in CO_2 emission reductions in Edinburgh.

One could extend the scenario generation and bike station location models to incorporate a range of socio-economic factors. We propose to leverage the Scottish Index of Multiple Deprivation (SIMD) for this extension. This data has employment statistics per data zone which can be used to generate more granular demand scenarios as inputs to the bike location model. Furthermore, SMID has crime statistics data, which could be used to prune bike location candidate nodes in areas of high crime, or incentivize bike rebalancing to areas of low crime outside of daylight hours. This is an important consideration as the previously unsuccessful scheme suffered from high crime-induced maintainability costs. Scalability of the bike location solution approach could be improved through alternative decomposition strategies. Firstly, one could introduce a node aggregation step which first aggregates nodes, solve an initial bike location problem on the aggregated nodes, prunes aggregated nodes not obtained in the solution, de-aggregate the selected nodes, and solve on the residual non-pruned nodes, as proposed in [15]. Secondly, one could explore the use of a Multi-Benders cut approach rather than using Single-Benders cuts and implement the algorithms using callbacks with Gurobi as we observed Gurobi significantly outperforming Xpress and the Benders Decomposition from CPLEX.

One could extend the modelling of our rebalancing solution approach to consider how an existing tour plan should be re-optimized. This will enable for re-optimization as bike levels change during the execution of the vehicle routes. Furthermore, one could relax the constraint that every node must be visited; and instead incentivize visiting each node through rebalancing incentives, in spirit of the Prize-Collecting TSP [16]. In this way, nodes are only visited if they are sufficiently useful in rebalancing. Finally, we recommend to explore the consideration of routing-costs when selecting bike locations. For this we refer the reader to a survey of recent developments in location-routing problems [17].

To improve the usability of the MIP-based approach in network rebalancing, one could warm-start the MIP with a feasible solution obtained by heuristic approaches, such as from our LKH-3 solution. Furthermore, one could explore decomposition approaches to enable the MIP for solving instances beyond 40 nodes. One could decompose networks into partitions of nodes; each which can be solved optimally. Furthermore, as only a subset of decision variables are typically used in the final VRP solution, column generation could be an effective approach to improve scalability of the MIP-based solution.

References

[1] Xinze Qiu, Tianli Gao, Yu Yang, Ankang Luo, Fan Shang, and Ruiqi Li. Understanding urban congestion with biking traffic and routing detour ratio, 2022.

[2] Gilbert Laporte, Stefan Nickel, and Francisco Saldanha-da Gama. *Location Science*. 01 2019.

[3] Richard Church and Charles R. Velle. The maximal covering location problem. *Papers in Regional Science*, 32(1):101–118, 1974.

[4] J.F. Benders. Partitioning procedures for solving mixed-variables programming problems. *Numerische Mathematik*, 4:238–252, 1962/63.

[5] R. M. Van Slyke and Roger Wets. L-shaped linear programs with applications to optimal control and stochastic programming. *SIAM Journal on Applied Mathematics*, 17(4):638–663, 1969.

[6] Bruno P. Bruck and Anand Subramanian. *Bike-Sharing Rebalancing Problems*, pages 1–9. Springer International Publishing, Cham, 2020.

[7] Keld Helsgaun. An effective implementation of the lin–kernighan traveling salesman heuristic. *European Journal of Operational Research*, 126(1):106–130, 2000.

[8] M. R. Rao. Technical note: A note on the multiple traveling salesmen problem. *Operations Research*, pages 628–632, June 1980.

[9] Roy Jonker and Ton Volgenant. Transforming asymmetric into symmetric traveling salesman problems: erratum. *Operations Research Letters*, 5(4):215–216, 1986.

[10] Scottish Government. Scottish index of multiple deprivation 2020. 2020.

[11] Ph. Blanchard D. Volchenkov. Markov chain methods for analyzing complex transport networks. *arXiv*, page 32210, 2012.

[12] S. N. Yaliraki M. Barahona M. T. Schaub, J.-C. Delvenne. Markov dynamics as a zooming lens for multiscale community detection: Non clique-like communities and the field-of-view limit. *PLoS ONE*, 1(e):32210, 2012.

[13] Martin Desrochers and Gilbert Laporte. Improvements and extensions to the miller-tucker-zemlin subtour elimination constraints. *Operations Research Letters*, 10(1):27–36, 1991.

[14] C. E. Miller, A. W. Tucker, and R. A. Zemlin. Integer programming formulation of traveling salesman problems. *J. ACM*, 7(4):326–329, October 1960.

[15] Ivona Gjeroska. *Solution methods for some variants of the vehicle routing problem*, chapter 4, pages 90–93. University of Edinburgh, 2023. PhD thesis.

[16] Egon Balas. The prize collecting traveling salesman problem. *Networks*, 19(6):621–636, 1989.

[17] Rossana Cavagnini, Alberto Santini, and Michael Schneider. Recent developments in location-routing problems: Deterministic single-echelon, single-objective, single-period problems. *European Journal of Operational Research*, 2025.

A Bike Rebalancing Visual Solutions



Figure 5: An optimal solution to the 20 node Bike Rebalancing Problem instance with 2 vans, solved by Xpress MIP.



Figure 6: An optimal solution to the 30 node Bike Rebalancing Problem instance with 2 vans, solved by Xpress.

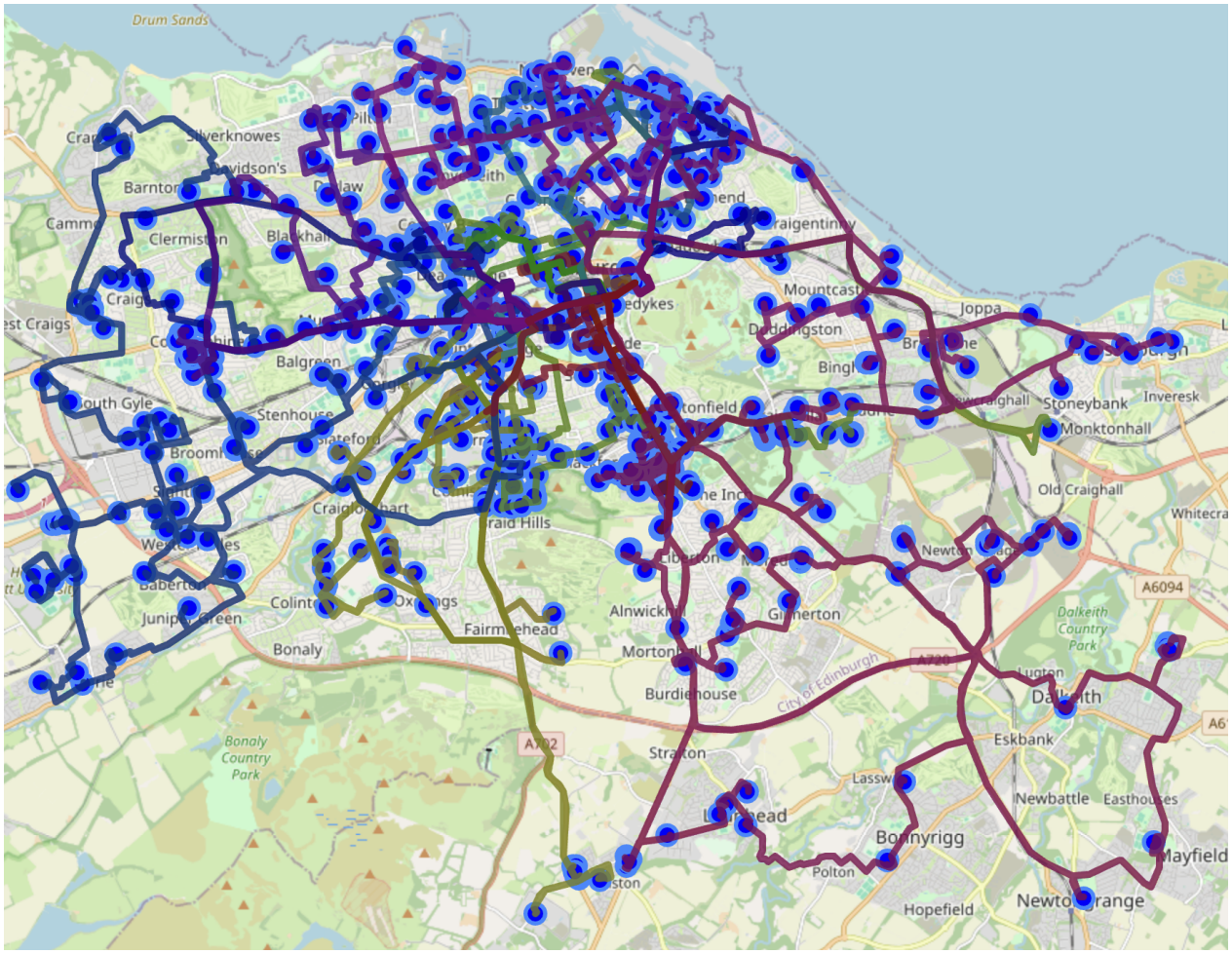


Figure 7: LKH-3 solution to the 1000 node Bike Rebalancing Problem with 50 vans after 1000 seconds.